



Addis Ababa University
College of Natural and Computational Sciences
Department of Computer Science

Course Title: Introduction to Emerging Technologies

Course code: EmTe 1012

ECTS: 5

Course description:

This course will enable students to explore current breakthrough technologies in the areas of Artificial Intelligence, Internet of Things and Augmented Reality that have emerged over the past few years. Besides helping learners become literate in emerging technologies, the course will prepare them to use technology in their respective professional preparations.

Course objective:

By the end of this course the student will be able to:

- Identify different emerging technologies
- Differentiate different emerging technologies
- Select appropriate technology and tools for a given task
- Identify necessary inputs for application of emerging technologies

Course outline:

Chapter 1: Introduction to Emerging Technologies

- Evolution of technologies
 - Introduction to Industrial revolution
 - Historical background (IR 1.0, IR 2.0, IR 3.0)
 - Fourth industrial revolution (IR 4.0)

- Role of data for Emerging technologies
- Enabling devices and networks for emerging technologies (programmable devices)
- Human to Machine Interaction
- Future trends in emerging technologies

Chapter 2: Introduction to Data Science

- Overview for Data Science
 - Definition of data and information
 - Data types and representation
- Data Value Chain
 - Data Acquisition
 - Data Analysis
 - Data Curating
 - Data Storage
 - Data Usage
- Basic concepts of Big data

Chapter 3: Artificial Intelligence

- Introduction to AI
 - What is AI
 - History of AI
 - Levels of AI
 - Types of AI
- Applications of AI
 - Agriculture
 - Health
 - Business (Emerging market)
 - Education
- AI tools and platforms (eg: scratch/object tracking)
- Sample application with hands on activity (simulation based)

Chapter 4: Internet of things (IoT)

- Overview of IOT
 - What is IOT?
 - History of IOT
 - Advantages of IOT
 - Challenges of IOT
- How IOT works
 - Architecture of IOT
 - Devices and network
- Applications of IOT
 - Smart home
 - Smart grid
 - Smart city
 - Wearable devices
 - Smart farming
- IOT tools and platforms (eg: KAA IoT /Device Hive/Zetta/Things Board...)
- Sample application with hands on activity (eg IOT based smart farming)

Chapter 5: Augmented Reality (AR)

- Introduction to AR
- Virtual reality (VR), Augmented Reality (AR) vs. mixed reality (MR)
- Architecture of AR systems.
- Application of AR systems (education, medical, assistance, entertainment) workshop oriented hands demo

Chapter 6: Ethics and Professionalism of Emerging Technology

- Technology and ethics
- Digital privacy
- Accountability and trust
- Treats and challenges

Chapter 7: Other Emerging Technologies

- Nanotechnology
- Biotechnology
- Blockchain technology
- Cloud and quantum computing
- Autonomic computing
- Computer vision
- Embed systems
- Cyber security
- Additive manufacturing (3D Printing)
- Etc. ...

Evaluation Schemes:

- Test 20%
- Quiz 10%
- Assignment 20%
- Final Examination 50%

References:

1. Follett, J. (2014). Designing for Emerging Technologies: UX for Genomics, Robotics, and the Internet of Things: O'Reilly Media.
2. Vong, J., & Song, I. (2014). Emerging Technologies for Emerging Markets: Springer Singapore.
3. Del Rosal, V. (2015). Disruption: Emerging Technologies and the Future of Work. Emtechub.
4. Sadiku, M. N. O. (2019). Emerging Internet-Based Technologies: CRC Press.
5. Mohamed Anis Bach Tobji, Rim Jallouli, Yamen Koubaa, Anton Nijholt Digital Economy. Emerging Technologies and Business Innovation, 2018.
6. Mahdi H. Miraz, Peter Excell, Andrew Ware, Safeeullah Soomro, Maaruf Ali, Emerging Technologies in Computing, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering 200, Springer International Publishing, 2018.
7. Francesco Corea. Artificial Intelligence and Exponential Technologies: Business Models Evolution and New Investment Opportunities, 2017.